NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

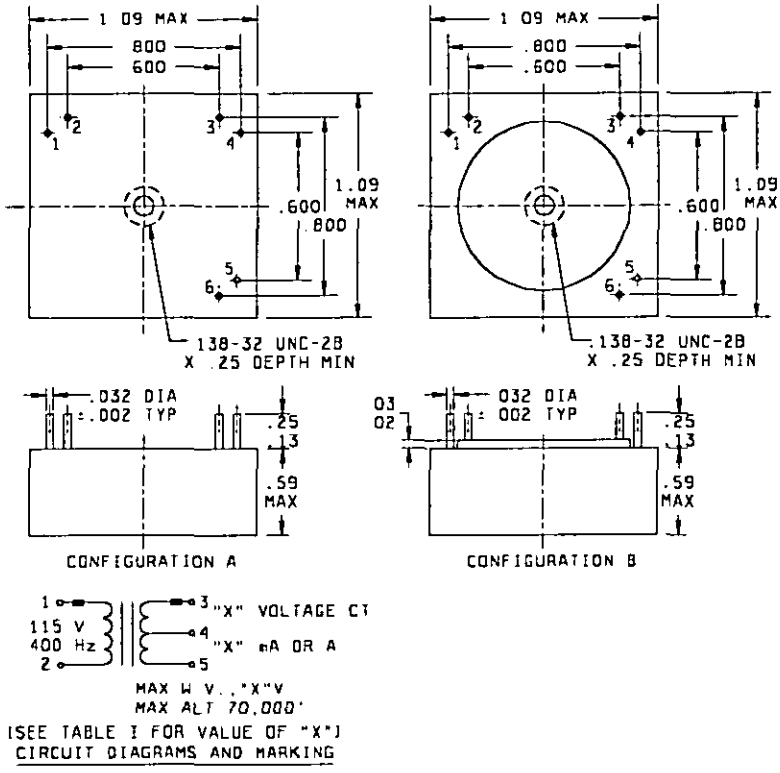
MIL-PRF-27/39B
7 February 1984
SUPERSEDING
MIL-T-27/39A
20 October 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 3 VOLT AMPERES, 400 HZ,
TF5S03ZZ488 TO TF5S03ZZ531, INCLUSIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.138	3.51
.25	6.4
.59	15.0
.600	15.24
.75	19.1
.800	20.32
1.09	27.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

(B) denotes changes

FSC 5950

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.1 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 3 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 75 percent minimum at rated load.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ shall not exceed 18 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 20°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

Ⓑ QUALITY ASSURANCE PROVISIONS:

Extent of qualification:

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-T-27/37 through MIL-T-27/47 inclusive, all parts.

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Qualification testing and approval to M27/39-44 shall be sufficient to grant qualification approval to M27/39-01 through M27/39-44.

Part number example:

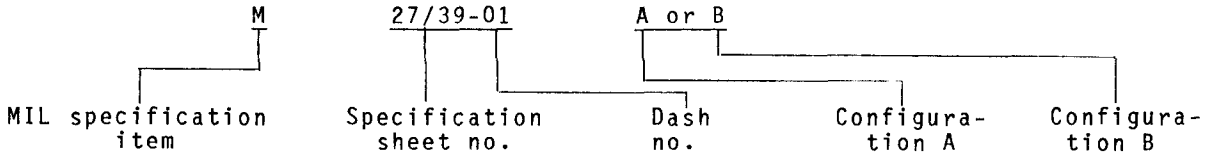


TABLE I. Electrical ratings.

Dash no.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	488	6	0.500	325
02 "	489	6.3	0.476	"
03 "	490	7	0.429	"
04 "	491	7.5	0.400	"
05 "	492	8	0.375	"
06 "	493	9	0.333	"
07 "	494	10	0.300	"
08 "	495	11	0.273	"
09 "	496	12	0.250	"
10 "	497	12.6	0.238	"
11 "	498	13	0.231	"
12 "	499	15	0.200	"
13 "	500	16	0.188	"
14 "	501	18	0.167	"
15 "	502	20	0.150	"
16 "	503	22	0.136	"
17 "	504	24	0.125	"
18 "	505	26	0.115	"
19 "	506	28	0.107	"
20 "	507	30	0.100	"
21 "	508	33	0.091	"
22 "	509	36	0.083	"
23 "	510	39	0.077	"
24 "	511	43	0.070	"
25 "	512	47	0.064	"
26 "	513	51	0.059	"
27 "	514	56	0.054	"
28 "	515	62	0.048	"
29 "	516	68	0.044	"
30 "	517	75	0.040	"
31 "	518	82	0.037	"
32 "	519	91	0.033	"
33 "	520	100	0.030	"
34 "	521	110	0.027	"
35 "	522	115	0.026	"
36 "	523	120	0.025	"
37 "	524	130	0.023	"
38 "	525	150	0.020	"
39 "	526	180	0.017	"
40 "	527	200	0.015	440
41 "	528	220	0.014	440
42 "	529	240	0.013	440
43 "	530	270	0.011	580
44 "	531	300	0.010	580

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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